EXP 7: Customer Order Analysis Using Pandas

AIM

To analyze customer orders using Pandas DataFrame operations.

ALGORITHM

1. Import Pandas.
2. Get customer IDs, order dates, product names, and quantities.
3. Create a DataFrame.
4. Calculate customer order counts and average product quantities.
5. Display the earliest and latest order dates.

SHORT CODE

EXP 7 – Customer Order Analysis

```
import pandas as pd


c = input("Customer IDs: ").split()
d = input("Order dates (YYYY-MM-DD): ").split()
p = input("Product names: ").split()
q = list(map(int, input("Quantities: ").split()))


df = pd.DataFrame({"Customer": c, "Date": pd.to_datetime(d),
                   "Product": p, "Quantity": q})


print(df.groupby("Customer").size())
print(df.groupby("Product")["Quantity"].mean())
print("Earliest:", df["Date"].min())
print("Latest:", df["Date"].max())
```

INPUT

Customer IDs: C1 C2 C1 C2

Order dates (YYYY-MM-DD): 2026-01-10 2026-01-12 2026-02-05 2026-02-08

Product names: Pen Book Pen Book

Quantities: 2 4 6 8

OUTPUT

Customer

C1 2

C2 2

Product

Book 6.0

Pen 4.0

Earliest: 2026-01-10

Latest: 2026-02-08

OUTPUT:

```
============ RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 7.py ======
Customer IDs: A1 E1 S1
Order dates (YYYY-MM-DD): 2026-02-23 2026-11-30 2026-11-04
Product names: PEN BOOK PENCIL
Quantities: 2 4 7
Customer
A1    1
E1    1
S1    1
dtype: int64
Product
BOOK      4.0
PEN       2.0
PENCIL    7.0
Name: Quantity, dtype: float64
Earliest: 2026-02-23 00:00:00
Latest: 2026-11-30 00:00:00
```

RESULT

Customer order counts, average quantity per product, and earliest and latest order dates were calculated successfully.